A Controlled Lensing/X-ray Cross-Readout Enrichment Audit in HFF and CLASH Clusters

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Abstract

We present a controlled X-ray/lensing enrichment audit for six Hubble Frontier Fields strong-lensing clusters and a frozen external CLASH map-level validation sample. For each system we compare radius- and exposure-controlled *Chandra* X-ray structure against public convergence and magnification maps. The primary statistic is top-region enrichment relative to matched null controls, used here as a cross-readout enrichment diagnostic. In the HFF development sample, all six clusters have median residual/null ratios above unity, and 88 of 92 tested model/observable/tracer rows are above their matched-null mean. In the external CLASH validation sample, 12 non-HFF clusters pass the frozen population endpoint: a spatial-block-aware hierarchical fit gives a population residual/null ratio of 1.231 with an approximate 95 percent interval of 1.119–1.353. However, weak and below-null magnification/residual channels remain in the row table. We therefore interpret the result as a model-sensitive observational enrichment statistic, not as a universal lensing-readout law, a final physical field, a derived mechanism, or a dark-matter replacement proof.

Contents

1 Introduction **2**
1.1 Source-transport boundary 2

2 Observational Hypothesis **2**

3 Data and Provenance **3**

4 Methods **3**

5 Results **4**

6 Negative Controls **4**

7 Statistical Hardening Status **4**

8 Compact-Source Masking **5**

9 Model-Family Uncertainty **5**

10 Conventional Explanations **5**

11 Candidate Visibility Maps **6**

12 Interpretation and Scope 7
13 Falsification Criteria 7
14 External Blind Protocol 7
15 HFF All-Six Expansion Status 8
16 External CLASH Map-Level Validation 8
17 Reproducibility Layer 9
18 First-Pass Hierarchical Meta-Analysis 9
18.1 Full-4D scoring boundary 10
19 Limitations 10
20 Conclusion 10

1 Introduction

Cluster lensing analyses compare several imperfect readouts of the same projected system: reconstructed convergence, magnification, X-ray structure, galaxy light, and model-family assumptions. This paper asks a deliberately narrow question about those readouts. It does not try to solve the mass-tracer gap by introducing a new hidden component. The useful question is narrower and falsifiable:

> Can controlled lensing/X-ray cross-readout enrichment be detected without adding a fitted hidden density component?

This draft studies that question using Hubble Frontier Fields lensing maps and *Chandra* X-ray structure. The goal is not to derive a physical mechanism. The goal is to define and test an observational prefilter:

> Do controlled X-ray residual structures preferentially lie inside high lensing-readout regions above matched geometry-preserving nulls?

1.1 Source-transport boundary

The present enrichment statistic compares terminal observational maps. It does not reconstruct a smooth optical source arrow, a body connection, or an observer–source transport law. In the current Tau Core source ledger, a non-scalar smooth optical transport can arise without a fitted gain only from one of two independently supplied structures: holonomy of a full body connection, or Kato transport of a varying, constant-rank, source-owned access projector, with generator $[dP_O, P_O]$. A scalar phase connection may carry clock or phase information, but it cannot determine normalized non-scalar optical orientation. Consequently, a positive value of Eq. (2) remains a cross-readout association and cannot by itself identify either optical source mechanism.

2 Observational Hypothesis

Let

$$X_{\text{res}} = \text{Chandra X-ray structure after radius and exposure controls} \tag{1}$$

and let L be a declared HFF lensing readout, either convergence κ or magnification μ. The cross-readout enrichment hypothesis predicts

$$V_{\text{obs}}(L, X_{\text{res}}) > V_{\text{null}}(L, X_{\text{res}}), \tag{2}$$

where V_{obs} is top-region enrichment and V_{null} is the matched-null expectation. Equation 2 is a screening diagnostic, not a field equation. It is only meaningful after declaring the cluster, model family, observable channel, tracer definition, and control family. This prevents the claim from becoming a post-hoc statement that any attractive overlap between any two maps supports a preferred physical theory.

3 Data and Provenance

The development analysis uses public HFF lens-model products and *Chandra* X-ray products. The lensing side includes the HFF model families available in the local input set: Merten, CATS, Diego, and Williams. The development cluster sample is Abell S1063, Abell 2744, Abell 370, MACS0416, MACS0717, and MACS J1149.5+2223.

The external validation sample uses public CLASH lensing products and archived *Chandra* data. Before inspecting the external endpoint, the branch froze a 12-cluster non-HFF CLASH sample: RX J1347.5-1145, MACS J2129.4-0741, MACS J1206.2-0847, CL J1226.9+3332, MS 2137.3-2353, MACS J1423.8+2404, Abell 2261, MACS J1311.0-0310, MACS J1931.8-2635, RX J1532.9+3021, MACS J0744.9+3927, and MACS J0329.7-0211. The external map-level runner uses Zitrin LTM-Gauss and Zitrin NFW CLASH convergence and magnification products, paired with uniform *Chandra* broad-band flux/exposure products reduced in the local CIAO pipeline [11, 12].

The HFF lens-model archive and published model-comparison references are treated as the current bibliography scaffold [6–10]. The software layer uses the Astropy ecosystem [1–3], reprojection tooling [4], and *Chandra*/CXC resources [5]. The local paper branch includes a README-level model-product audit covering the method-family citation keys for SaWLens, LENSTOOL/CATS, WSLAP+, and GRALE. Before submission, those method references must receive a final publisher/ADS metadata check and exact acknowledgement text.

4 Methods

The paper-facing controlled tracer is

$$X_{\text{res}} = \texttt{chandra\_xray\_radius\_exposure\_residual\_z}. \tag{3}$$

For each cluster, model family, and lensing observable, the runner measures

$$E_{\text{obs}} = \frac{\text{fraction(top } L \text{ inside top } X_{\text{res}})}{\text{base fraction(top } X_{\text{res}})}. \tag{4}$$

The paper-facing diagnostic ratio is

$$R_{\text{null}} = \frac{E_{\text{obs}}}{\langle E_{\text{null}} \rangle}. \tag{5}$$

Values above unity indicate excess relative to the declared null. Values below unity are retained as weak or null channels.

5 Results

Across the all-six first-pass runner table, 92 model/observable/tracer rows are tested. Of these, 88 are above matched null and four are below matched null. At cluster level, all six systems have median residual/null ratios above unity: Abell S1063, Abell 2744, Abell 370, MACS0416, MACS0717, and MACS J1149.5+2223. Four clusters have every tested row above unity; Abell 370 and MACS0717 retain channel-level falsifiers.

The retained below-null rows are:

Cluster	Model	Observable	Tracer	$R_{\rm null}$
Abell 370	Williams v4.1	magnification	X-ray σ_1	0.744
MACS0717	Williams v4.1	magnification	X-ray σ_1	0.797
MACS0717	Williams v4.1	magnification	X-ray residual	0.873
MACS0717	Diego v4.1	magnification	X-ray σ_1	0.894

In the original four-cluster residual-only table, the below-null channel is:

$$E_{\rm obs} = 0.945, \tag{6}$$
$$\langle E_{\rm null}\rangle = 1.083, \tag{7}$$
$$R_{\rm null} = 0.873, \tag{8}$$

for MACS0717 Williams magnification, with empirical $p(\geq E_{\rm obs}) = 1.000$. This is not a nuisance row to hide. It is an internal falsifier.

6 Negative Controls

The negative-control summary uses roll and value-shuffle controls. The control ratios are generally above unity, supporting the interpretation that the signal is not merely a baseline top-region occupancy effect. Weak control rows remain and should be preserved as scope limits, especially for MACS0717 magnification and selected Abell 2744 control minima.

7 Statistical Hardening Status

The current statistics support an exploratory enrichment screen, not a population-level detection. In the all-six first-pass runner table, the matched-null summaries use 1000 null samples per row. Of 92 rows, 88 are above the matched-null mean and 88 are above the matched-null 95th percentile. The four retained below-null rows are all magnification-channel failures: one in Abell 370 and three in MACS0717. This is a model/readout sensitivity result, not a row to hide.

This is stronger than a visual-overlap claim, but it is still not a complete publication-grade uncertainty model. The next statistical hardening layer must add bootstrap confidence intervals on effect sizes, uncertainty bands on $R_{\rm null}$, spatial-autocorrelation-aware resampling, grouped multiple-hypothesis correction by cluster/model/channel family, and an explicit treatment of lens-model reconstruction covariance. Until those are complete, the correct language is exploratory enrichment screen rather than strong statistical detection.

The spatial-autocorrelation issue is the most important remaining technical limitation. Cluster maps are smoothed and neighboring pixels are not independent, so the effective degrees of freedom can be much smaller than the pixel count. A publication-grade detection claim would require at minimum a spatial block bootstrap, annulus-preserving block shuffles, a correlation-length sweep, and a cluster-level meta-analysis in which the six clusters, not the pixels, are the independent units.

At the current cluster level, all six HFF systems have median cross-readout enrichment above matched null. Four of the six clusters have all tested rows above unity; Abell 370 and MACS0717 retain specific below-null magnification channels. This cluster-level summary is more appropriate than treating all pixels or rows as independent detections, but it does not overcome the small-sample limitation.

8 Compact-Source Masking

Compact-source masking addresses a simple contamination explanation: that the visibility signal is driven by a small number of compact X-ray peaks rather than cluster-scale morphology. For the original four-cluster audit we compare the unmasked X-ray/lensing enrichment to a strict compact mask ("compact p99, dilation 2"). The covered rows all survive this stress test: the minimum masked/unmasked enrichment ratio is 1.018. MACS0717 additionally retains annulus-preserving radial-null support under the same strict mask.

9 Model-Family Uncertainty

The candidate-map weight is a mean of independent HFF kappa percentile-rank maps. A mean map is useful, but it can hide disagreement among lens-model families. We therefore generate a companion uncertainty layer: mean kappa-rank, pixelwise kappa-rank scatter, and the fraction of model families that place a pixel in their top-rank support. The median cluster-level kappa-rank scatter is 0.186, and the largest cluster-level 90th-percentile scatter is 0.382. MACS0717 has the largest median scatter among the original four systems, consistent with the channel sensitivity already exposed by the Williams magnification falsifier.

10 Conventional Explanations

A positive X-ray/lensing enrichment pattern is not automatically evidence for a new physical mechanism. Several conventional explanations can produce co-location between X-ray residual structure and lensing readouts. Cluster merger geometry can align shocked or displaced gas with projected potential features. Baryon–potential alignment can make X-ray morphology and convergence high points trace the same underlying cluster structure. Hydrostatic or quasi-hydrostatic gas structure can correlate X-ray brightness with the gravitational potential even when the system is not fully relaxed. More generally, X-ray and lensing maps are both expected to co-trace aspects of the same projected mass distribution.

The lensing products themselves also introduce baseline explanations. Reconstruction covariance, smoothing scale, model-family priors, and mass-concentration coupling can move high-rank convergence or magnification support toward the same cluster-scale features. These are not nuisance details; they are part of the null hypothesis that the audit must compete against.

Baseline	What it can explain	What remains after this audit
Cluster merger geometry	Projected merger axes can align gas structure, galaxies, and lensing support.	The statistic is still tested against radius/exposure and matched spatial nulls, but merger state is not yet an explicit covariate.
Baryon–potential co-tracing	Gas and lensing can both follow the same large-scale potential.	This may explain part of the positive enrichment; the audit asks whether the top-region excess survives declared controls, not whether co-tracing exists.
Hydrostatic or quasi-hydrostatic structure	X-ray brightness can correlate with the gravitational potential in relaxed or partly relaxed systems.	The residual tracer subtracts radius/exposure structure, but it does not yet fit a full thermodynamic cluster model.
Reconstruction covariance and smoothing	Lens-model regularization can broaden or shift high-rank support toward common cluster-scale features.	Model-family and channel sensitivity are retained explicitly; this is why below-null magnification rows are not removed.
Source masking and compact peaks	Point sources or compact gas peaks can create artificial top-region overlap.	Compact-source checks reduce the simplest contamination story, but a uniform public mask release remains future work.

The present analysis addresses only part of this conventional baseline. Radius and exposure controls reduce simple radial and instrumental explanations. Roll/value-shuffle controls reduce generic top-region occupancy explanations. Compact-source masking reduces the simplest X-ray point-peak contamination story. Model-family scatter exposes where the lensing readout is unstable. However, the analysis does not yet fully model merger state, hydrostatic structure, lensing reconstruction covariance, or smoothing-induced alignment. For this reason the result should be read as a controlled enrichment pattern that survives several simple baselines, not as proof that conventional cluster morphology has been exhausted.

This framing also defines what would make the result less interesting. If a future baseline model using merger-state covariates, gas thermodynamic structure, and lensing reconstruction covariance reproduces the same cluster-level and channel-level enrichment pattern, then the present statistic should be interpreted as a compact morphology diagnostic rather than evidence for a new physical readout. Conversely, if the enrichment remains positive after those baselines are modeled and preregistered on a larger external sample, then it becomes a stronger constraint on any later physical interpretation. The current paper deliberately stops at the first statement: the enrichment survives the controls implemented here, but conventional cluster morphology remains an active competing explanation.

11 Candidate Visibility Maps

Candidate maps were generated as proxy products:

$$T_{\text{Chandra}} = \max\left(0, \text{robust } z(\text{Chandra residual after radius/exposure grouping})\right), \tag{9}$$

and

$$T_{\text{vis}} = T_{\text{Chandra}} \left\langle \text{percentile rank}(\kappa_{\text{HFF}}) \right\rangle. \tag{10}$$

12 Interpretation and Scope

The current result supports an exploratory cross-readout enrichment result:

> controlled X-ray residual geometry is usually enriched inside high lensing-readout regions above matched nulls.

It does not support a universal lensing-readout claim, because Abell 370 and MACS0717 retain below-null magnification channels. It also does not require introducing a hidden component inside the observational statistic. The allowed interpretation in this paper is deliberately conservative: a controlled phenomenological enrichment statistic in cluster lensing. The disallowed interpretation is a renamed hidden density or a derived new-gravity mechanism.

In the current Tau Core foundation architecture, lensing and X-ray observables would be typed descendants of one shared preterminal descriptor only if each is constant on that descriptor's fibres. This paper neither constructs that descriptor nor tests the required no-bypass condition. Cross-readout enrichment is therefore compatible with, but does not establish, common parent-readout ancestry, terminal completeness or physical source occupation.

Projection-based frameworks, including τ-core-like interpretations, may treat this prefilter as an empirical constraint. They are not part of the paper's central claim. In particular, this paper does not provide a dynamical equation, action, field equation, relativistic consistency proof, lensing derivation, or covariant structure. Those belong to a separate theory paper.

13 Falsification Criteria

The branch would fail if:

1. the signal disappears under radius/exposure controls;
2. the signal is reproduced by roll or value-shuffle controls;
3. the signal depends on a single model family;
4. the analysis switches from magnification to convergence only after seeing the data;
5. a proposed physical interpretation predicts all Abell 370 or MACS0717 lensing products are positive.

The last condition is already active: Abell 370 and MACS0717 magnification failures forbid the strongest universal version of the claim.

14 External Blind Protocol

The next sample-expansion branch should be run under a frozen protocol. Before examining the enrichment outcome for any new cluster sample, the analysis must declare the cluster list, admissible model families, observable channels, primary tracer, compact-source mask, enrichment metric, top-region thresholds, null families, primary endpoint, secondary endpoints, and failure criteria. The primary unit should be the cluster, not the pixel or model/observable row. The primary endpoint should be the cluster median residual/null ratio across predeclared admissible rows. Kappa and magnification must remain separate channels, and a below-null channel must stay in the table rather than being replaced by a stronger observable after inspection.

This blind protocol is required before any population-level claim. In the current branch it has been applied once to the external CLASH map-level sample. The frozen external pass rule is that the spatial-block-aware hierarchical population residual/null ratio must have a lower approximate 95 percent interval above unity. This creates an external validation layer, while still requiring broader public replication before a population-level astrophysical claim.

15 HFF All-Six Expansion Status

The nearest homogeneous map-level expansion was to complete the two remaining HFF clusters, Abell 370 and MACS J1149.5+2223. In the current branch, the public lensing side of that extension has been acquired and audited: Merten, CATS, Diego, and Williams kappa and $z = 4$ magnification products are available locally for both clusters, with README files, checksums, finite-pixel checks, and WCS checks recorded in the acquisition manifest.

The *Chandra* side now has first-pass uniform products for both systems. A Chandra Data Archive TAP catalog query finds archived, no-grating observations for both systems: Abell 370 has 104 archived observations totaling 1998.062 ks, while MACS J1149.5+2223 has 8 archived observations totaling 365.425 ks. Both stacks were processed through the local CIAO pipeline and then through the common prefilter runner. This upgrades the branch from a four-cluster pilot to an all-six first-pass HFF audit, while retaining the need for spatial-autocorrelation-aware uncertainty and formal cluster-level meta-analysis before any population claim.

16 External CLASH Map-Level Validation

After the HFF all-six audit, the branch froze a direct external CLASH map-level validation run. The external runner does not pool CLASH with the HFF development rows. It repeats the same observational question on independent public cluster systems, with declared CLASH model families, declared lensing observables, declared *Chandra* tracers, and the same matched-null enrichment ratio.

The frozen CLASH run contains 96 model/observable/tracer rows across 12 clusters. Of these, 85 rows have residual/null ratio above unity. The row-level median is 1.205, with a minimum of 0.453 and a maximum of 1.757. The weakest rows are concentrated mainly in magnification plus residual-tracer channels, so the external validation preserves the model-sensitivity already visible in the HFF development audit.

Cluster	Median $R_{\rm null}$
MACS J0744.9+3927	1.113
MACS J0329.7-0211	1.113
CL J1226.9+3332	1.128
MACS J1206.2-0847	1.150
Abell 2261	1.152
MACS J1311.0-0310	1.167
MS 2137.3-2353	1.181
MACS J2129.4-0741	1.210
RX J1347.5-1145	1.217
MACS J1423.8+2404	1.220
MACS J1931.8-2635	1.309
RX J1532.9+3021	1.397

Spatial-block bootstrap uncertainties were then propagated into an external hierarchical fit with cluster and channel random intercepts. The fitted population log-ratio is

$$\hat{\mu}_{\rm CLASH} = 0.208,$$

corresponding to a population residual/null ratio of

$$\exp(\hat{\mu}_{\rm CLASH}) = 1.231,$$

with an approximate 95 percent interval of 1.119 to 1.353. The frozen external pass criterion is therefore satisfied. The fitted variance components are

$$\sigma_{\text{cluster}} = 0.076, \qquad \sigma_{\text{channel}} = 0.118, \qquad \sigma_{\text{residual}} = 0.026,$$

with median spatial-block log uncertainty 0.103.

This is stronger than the earlier proxy-level CLASH check because it uses actual map-level lensing/X-ray co-structure. It is still not a mechanism derivation and not a hidden-mass claim. The paper-safe reading is narrower: the same controlled cross-readout enrichment statistic that was developed on HFF clusters also passes a frozen external map-level CLASH validation.

17 Reproducibility Layer

The current branch now has an internal open-science scaffold. The frozen protocol YAML freezes the random seed, top-region thresholds, matched-null count, negative-control count, radius/exposure binning, compact-source mask, primary endpoint, secondary endpoints, and failure rules. A release manifest lists the protocol file, paper source, runner scripts, validators, generated tables, and figures that should be released together. The exact local files are the protocol YAML and reproducibility manifest listed in the project README and readiness ledger.

This does not yet make the branch fully public-reproducible. A public package still needs source-data download scripts, released masks, exact source-data metadata, final license/citation text, and one-command reproduction instructions. The protocol file is therefore an internal freeze for the present branch and a template for the prospective homogeneous sample expansion.

18 First-Pass Hierarchical Meta-Analysis

We fit a first-pass hierarchical model to the all-six HFF runner table. The response is

$$\log R_{ij} = \log \left(E_{\text{obs},ij} / \langle E_{\text{null},ij} \rangle \right),$$

with a population intercept, cluster random intercepts, model/channel random intercepts, and a residual term:

$$\log R_{ij} = \mu + a_{\text{cluster}[i]} + b_{\text{channel}[j]} + \epsilon_{ij}.$$

The spatial-block-aware fitted population mean is positive:

$$\hat{\mu} = 0.489, \qquad \exp(\hat{\mu}) = 1.630,$$

with an approximate 95 percent interval of 1.429 to 1.860 on the population residual/null ratio. The fitted variance components are

$$\sigma_{\text{cluster}} = 0.091, \qquad \sigma_{\text{channel}} = 0.246, \qquad \sigma_{\text{residual}} = 0.161.$$

The median spatial-block log uncertainty supplied to the fit is 0.114, with a maximum of 0.537.

A correlation-length calibration estimates the $1/e$ autocorrelation length of the X-ray tracer maps. The median estimated correlation length is 102 native pixels, with a 90th percentile of 225 pixels; the smoothed X-ray σ_1 maps have the longest tails. A block-size sensitivity sweep with target block-axis counts of 4, 6, 8, 12, 16, 24, and 32 gives stable population residual/null ratios from 1.611 to 1.633. This reduces the risk that the fitted population effect is an artifact of one arbitrary spatial grid, while retaining the caveat that very smooth tracer maps require conservative block choices.

This fit is useful because it quantifies a point already visible in the descriptive tables: channel variation is larger than cluster variation. The weakness is therefore not that whole clusters fail the audit, but that some lensing/readout channels, especially magnification channels, sit below the all-six pattern.

This is a real hierarchical fit with row-level spatial-block uncertainty, but it remains a development-sample fit. The external CLASH map-level analysis now provides the first frozen validation layer and should be reported separately from the HFF development rows. The two samples should not be pooled in one undifferentiated model until a broader, predeclared multi-sample hierarchy is specified.

18.1 Full-4D scoring boundary

The later observer-specific full-4D compiler does not alter the enrichment statistics reported here. A future Tau metric candidate must compare the complete descended Hessian with a standard same-support metric through

$$E_K = (K_{HH} - K_{\rm std}) - C K_{VV}^{-1} C^{\dagger},$$

not score the positive Schur term by itself. Its lensing and X-ray terminal images must be frozen upstream, exact duplicate information removed, and the full reconstruction and cross-readout covariance retained in one held-out likelihood. This manuscript provides an observational pre-filter, not such a source-owned metric packet or fixed Tau prediction.

19 Limitations

This draft is not submission-ready. The model-team README audit, method-family BibTeX key coverage, compact-source masking audit, and model-family uncertainty layer are complete at the current preprint-draft level. Remaining blockers are narrower: the bibliography still needs final publisher/ADS metadata verification; the HFF development sample remains small and HFF-selected, with only six clusters; the two newly reduced HFF clusters are first-pass products; the CLASH external validation uses one public model-team family in two parameterizations and therefore does not replace broader multi-team external lens-model validation; MACS0717 has the strongest compact-source radial-null layer while the other clusters currently use enrichment-comparison masking audits; the first-pass random-effects meta-analysis now includes spatial-block row uncertainties, block-size sweep, correlation-length calibration, and one frozen external CLASH application, but the rule should still be preregistered and repeated on broader samples; the public reproducibility package is scaffolded but not yet released; the rotation-curve branch remains separate; and any projection-based physical interpretation, including τ-core-like interpretations, remains a later theory object until derived from explicit dynamical assumptions.

20 Conclusion

The HFF plus external CLASH branch supports a paper-worthy observational prefilter:

> controlled lensing/X-ray cross-readout enrichment is detectable as a model-sensitive spatial statistic in cluster lensing.

The correct claim is deliberately narrow: a falsifiable, map-level cluster-lensing enrichment statistic now survives both a HFF development audit and a frozen external CLASH validation. Its main value is as an empirical phenomenology and as a constraint on later physical interpretations, not as a derivation of such an interpretation.

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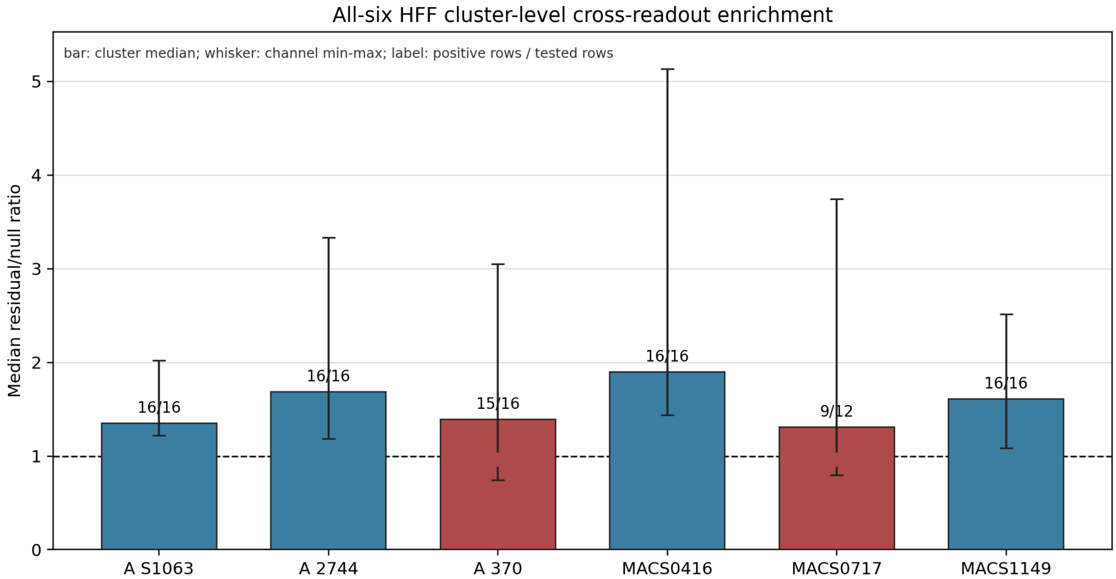

Figure 1: All-six HFF cluster-level cross-readout enrichment summary. Bars show the median observed/matched-null ratio across predeclared model/observable/tracer rows for each cluster; whiskers show the within-cluster channel minimum and maximum. Text labels report positive rows over tested rows. Red markers indicate clusters with retained below-null channels. This is the preferred paper-facing summary because the cluster, not the pixel or row, is the exploratory unit.

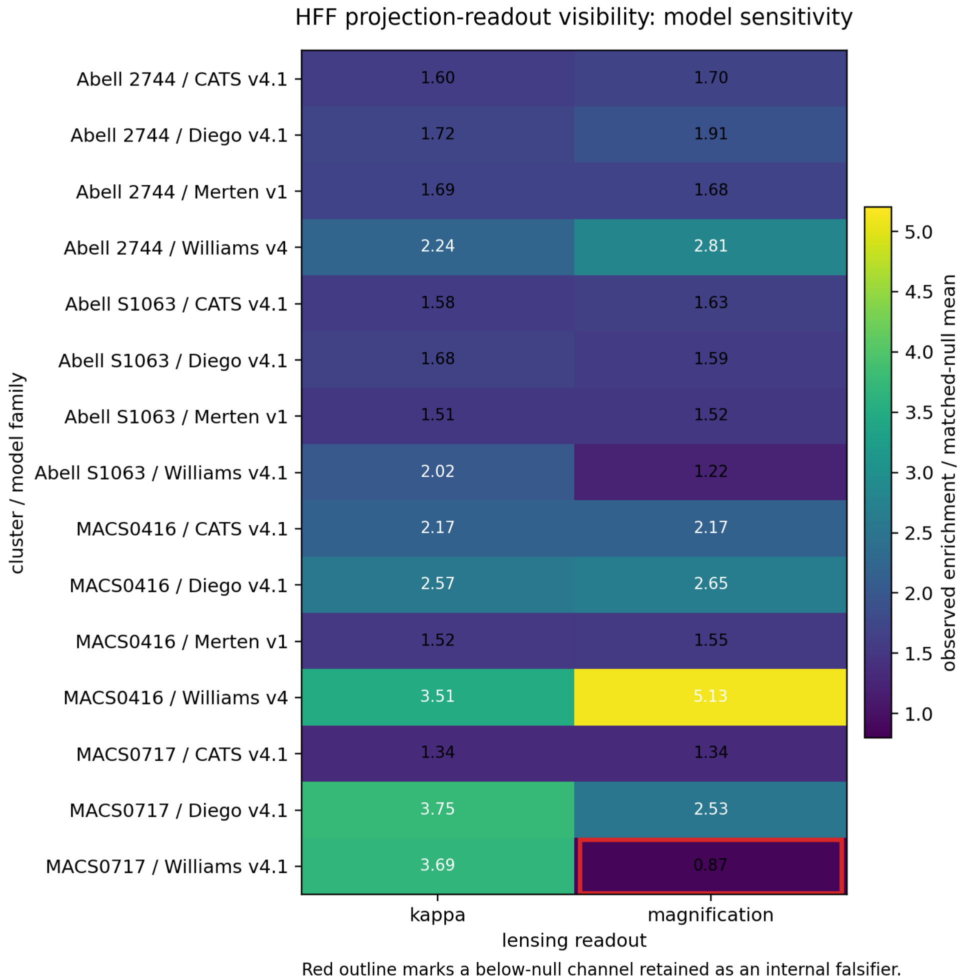

Figure 2: Model-sensitivity of HFF cross-readout enrichment. Each entry reports the ratio between observed enrichment and the matched-null mean for the controlled *Chandra* radius+exposure residual tracer in the original four-cluster paper-facing sensitivity table. Values above unity indicate excess co-location relative to the declared null. The MACS0717 Williams magnification entry is below unity and is retained as an internal falsifier.

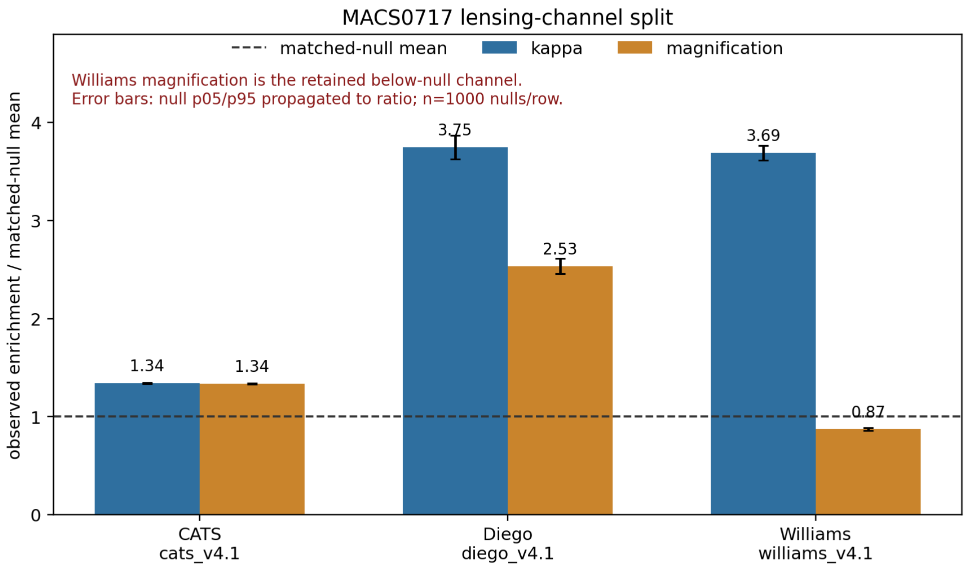

Figure 3: MACS0717 lensing-channel split. CATS and Diego are positive in both convergence and magnification. Williams convergence is positive, while Williams magnification falls below matched null. Error bars propagate the matched-null p05/p95 interval to the displayed observed/null ratio, with $n = 1000$ null samples per row. This demonstrates that the observed enrichment pattern is channel-sensitive and cannot be stated as a universal property of all lensing products.

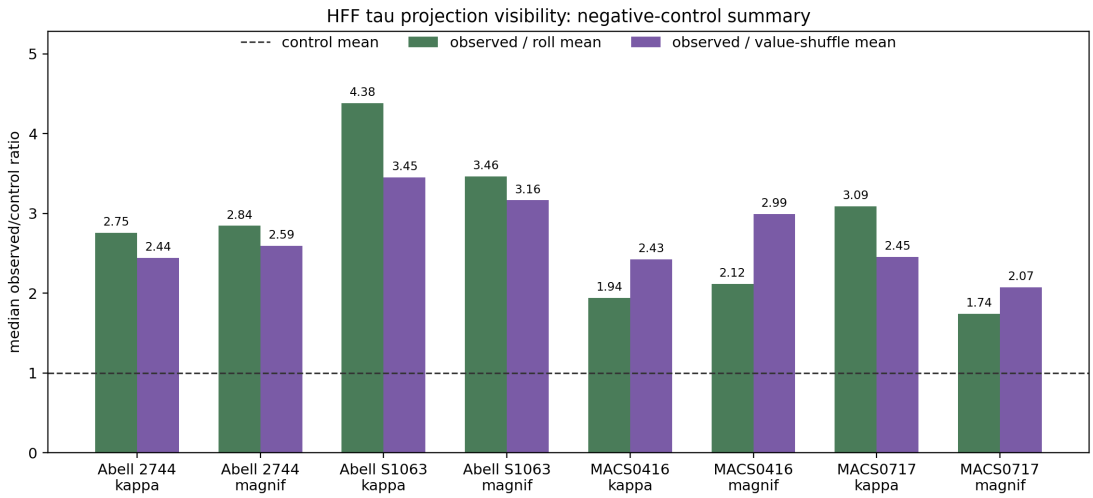

Figure 4: Negative-control summary. Bars show median observed/control ratios for roll and value-shuffle controls, grouped by cluster and lensing observable. The dashed line marks the control mean.

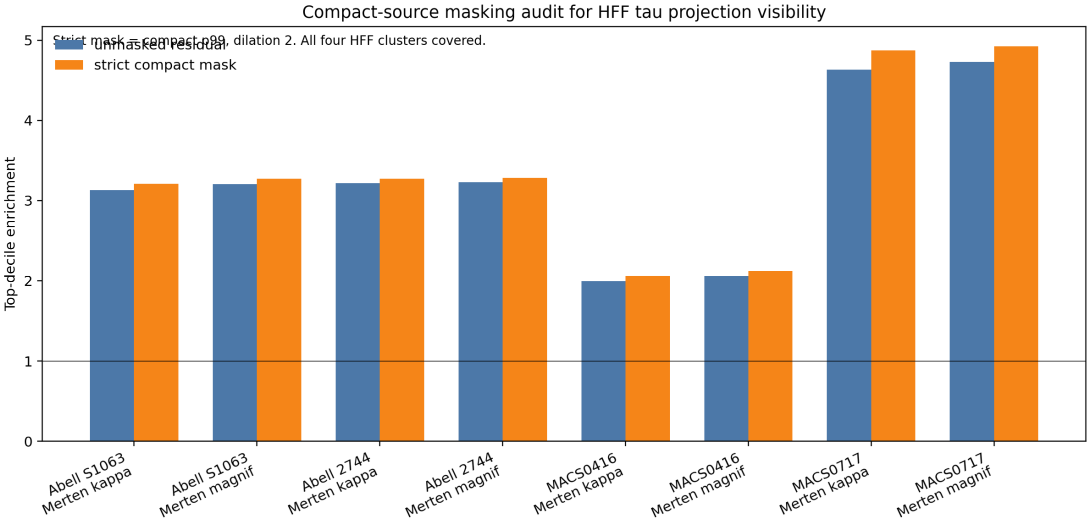

Figure 5: Compact-source masking audit. Bars compare unmasked and strict compact-masked X-ray/lensing enrichment for the covered clusters. The test does not prove a physical mechanism, but it argues against the simplest compact X-ray-peak contamination explanation for the enrichment statistic.

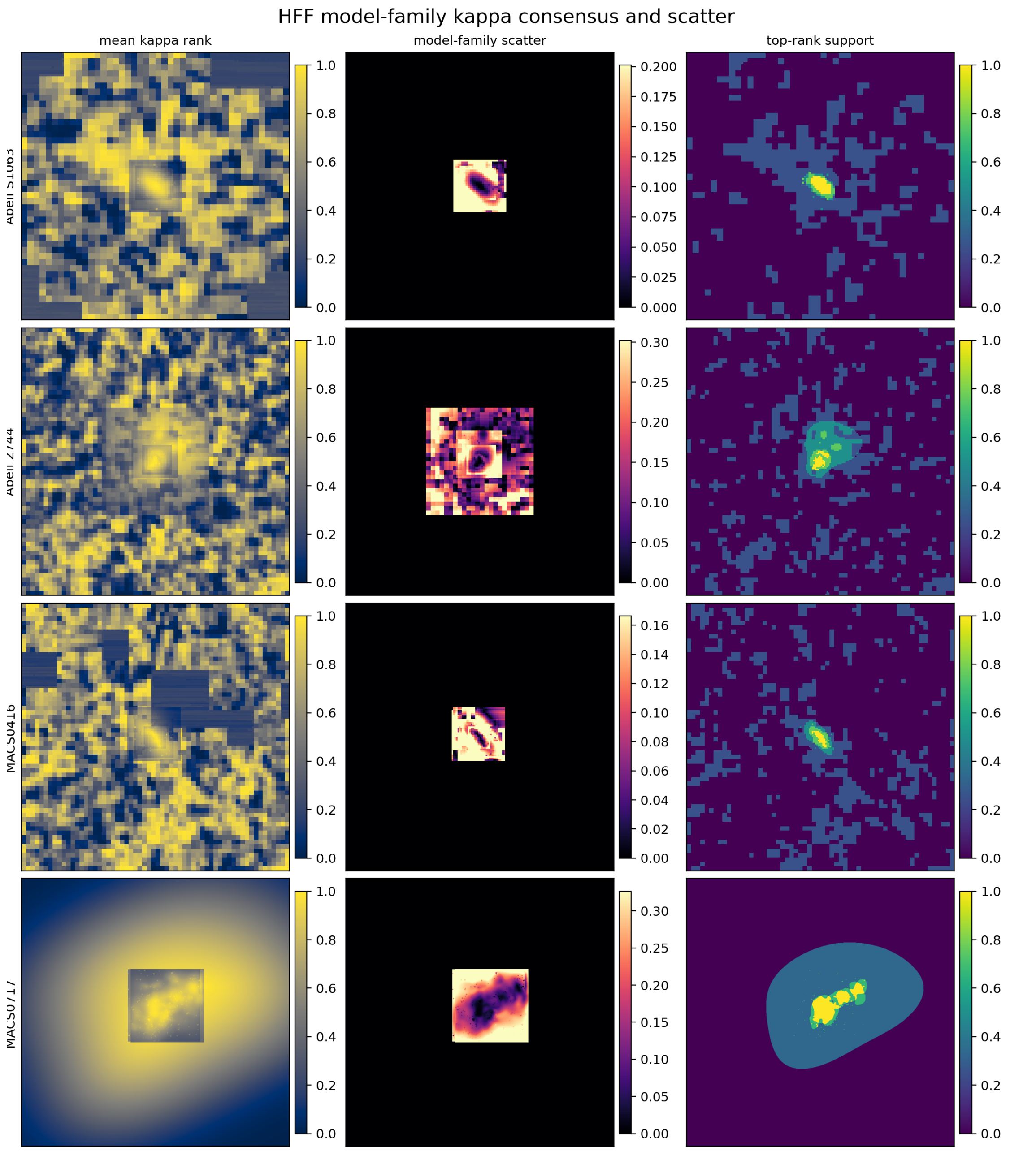

Figure 6: Model-family kappa consensus and scatter. The left column shows mean kappa percentile-rank support, the middle column shows pixelwise scatter across available model families, and the right column shows top-rank support fraction. This panel is an honesty layer for the candidate visibility maps: high mean support should be read together with model-family scatter.

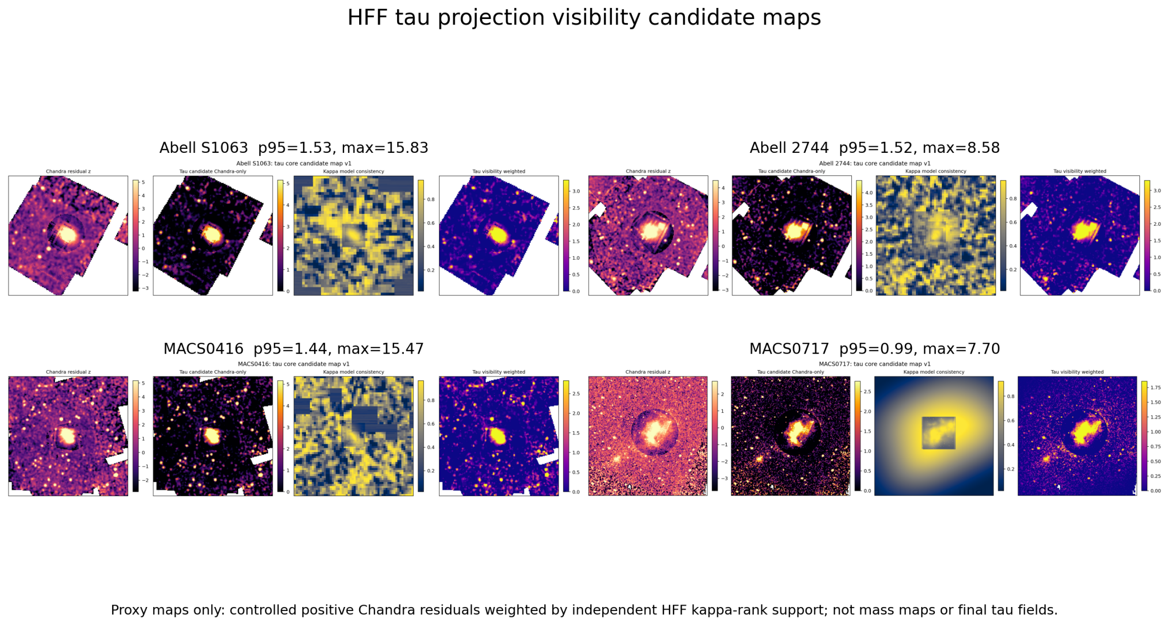

Figure 7: Candidate visibility maps for the original four HFF clusters. The displayed field is not a mass map, a final physical field, or a derived mechanism. It is a controlled positive *Chandra* residual weighted by the percentile-rank support of independent HFF convergence maps.

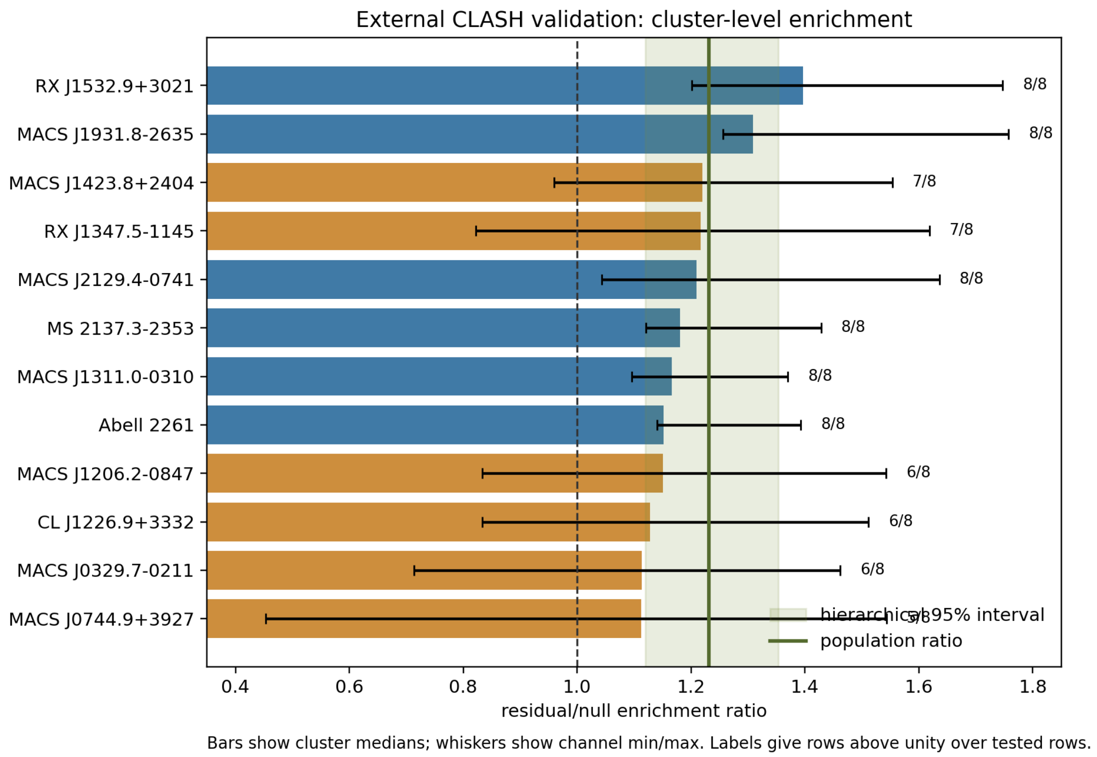

Figure 8: External CLASH map-level validation. Bars show cluster-level median residual/null enrichment ratios across the frozen Zitrin LTM-Gauss/NFW kappa/magnification and *Chandra* tracer rows; whiskers show the within-cluster channel minimum and maximum. The shaded band shows the approximate 95 percent hierarchical population interval, and the vertical green line shows the fitted population ratio. Labels report rows above unity over tested rows.

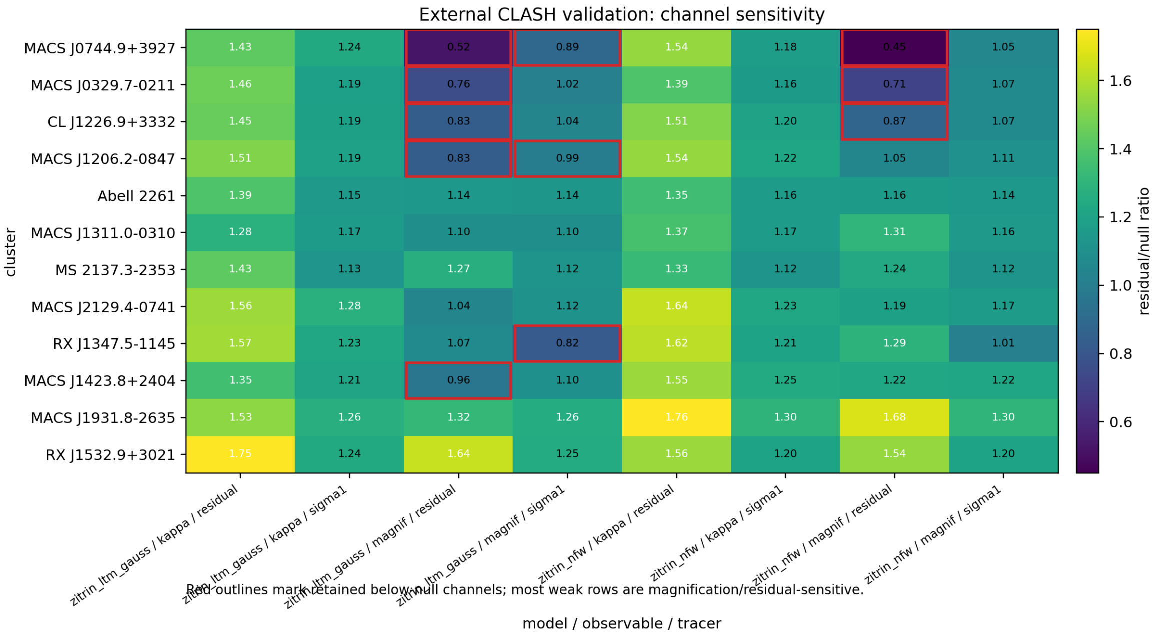

Figure 9: External CLASH channel sensitivity. Each cell is the observed enrichment divided by the matched-null mean for one frozen cluster/model/observable/tracer row. Red outlines mark retained below-null channels. The figure is intentionally not collapsed to a single positive number: the external result passes at population level while preserving magnification/tracer sensitivity as a real limitation.